



GE Medical Systems

Technical Publications

**Direction 2354338
Revision 0**

Signa LX Release 9.0 Patch 4N SRFD II Software Load Instructions

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Immediately complete a "Damage Loss Claim Form", available via MS Exchange Mail, after the damage is found.

MS Exchange Path:

Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/DAMAGE LOSS CLAIM FORM.

Send the completed form to the email address listed in the form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address (case sensitive):

<ftp://3.87.40.2/globepro/qualsys/Docs/190016MF.PDF>

Rev. 11/15/2000

REVISION HISTORY

<u>REV</u>	<u>DATE</u>	<u>PRIMARY REASON FOR CHANGE</u>
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0	Sept. 6, 2002	Initial version of procedure.
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- This revision/letter corresponds to the indicated document's revision control system.

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1- OVERVIEW

This CD-ROM, 2350782, contains the software patch to support SRFD2 hardware for the Signa Lx Release 9.0 M4 product. The CD ROM is structured as part of the 2156158-28 (software collector for Release 9.0).

Note

Signa software must be at the M4 version of Release 9.0 to install this patch CD. During the installation of the patches, the system must be idle and rebooted immediately after the installation.

2- REQUIRED PARTS

2-1 Additional parts

- Patch CD 2350782 Rev: SW001, Patch name list: patch_90_M4_04N
- Tech Pub 2354338 Rev 0 Signa Lx Release 9.0 Patch 4N SRFDII Software Load Instructions

IMPORTANT!

THE SIGNA LX RELEASE 9.0 PATCH 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 17 CD, 2327548-3 REV 004 AND TECH PUB 2329420 REV 2 SIGNA LX RELEASE 9.0 PATCH SOFTWARE LOAD INSTRUCTIONS MUST BE LOADED PRIOR TO INSTALLING PATCH 4N, WHICH IS DETAILED IN THIS DIRECTION.

1. Prior to loading the Patch CD, determine which patches have already been loaded on your system.
2. Inside a C Shell window, type the following

```
cd /usr/g/bin/ <enter>
```

```
getver <enter>
```

Build number for patch_90_M4_04 is 90.29F_M4_0139.z.PJ

Build number for patch_90_M4_05 is 90.29F_M4_0139.z.PH

Build number for patch_90_M4_06 is 90.29F_M4_0139.z.PG

Build number for patch_90_M4_07 is 90.29F_M4_0139.z.PL

Build number for patch_90_M4_08 is 90.29F_M4_0139.z.PH

Build number for patch_90_M4_09 is 90.29F_M4_0139.z.PD

Build number for patch_90_M4_11 is 90.29F_M4_0139.z.PG

Build number for patch_90_M4_12 is 90.29F_M4_0139.z.PK

Build number for patch_90_M4_13 is 90.29F_M4_0139.z.PJ

Build number for patch_90_M4_14 is 90.29F_M4_0139.z.PL

Build number for patch_90_M4_15 is 90.29F_M4_0139.z.PJ

Build number for patch_90_M4_17 is 90.29F_M4_0139.z.PJ

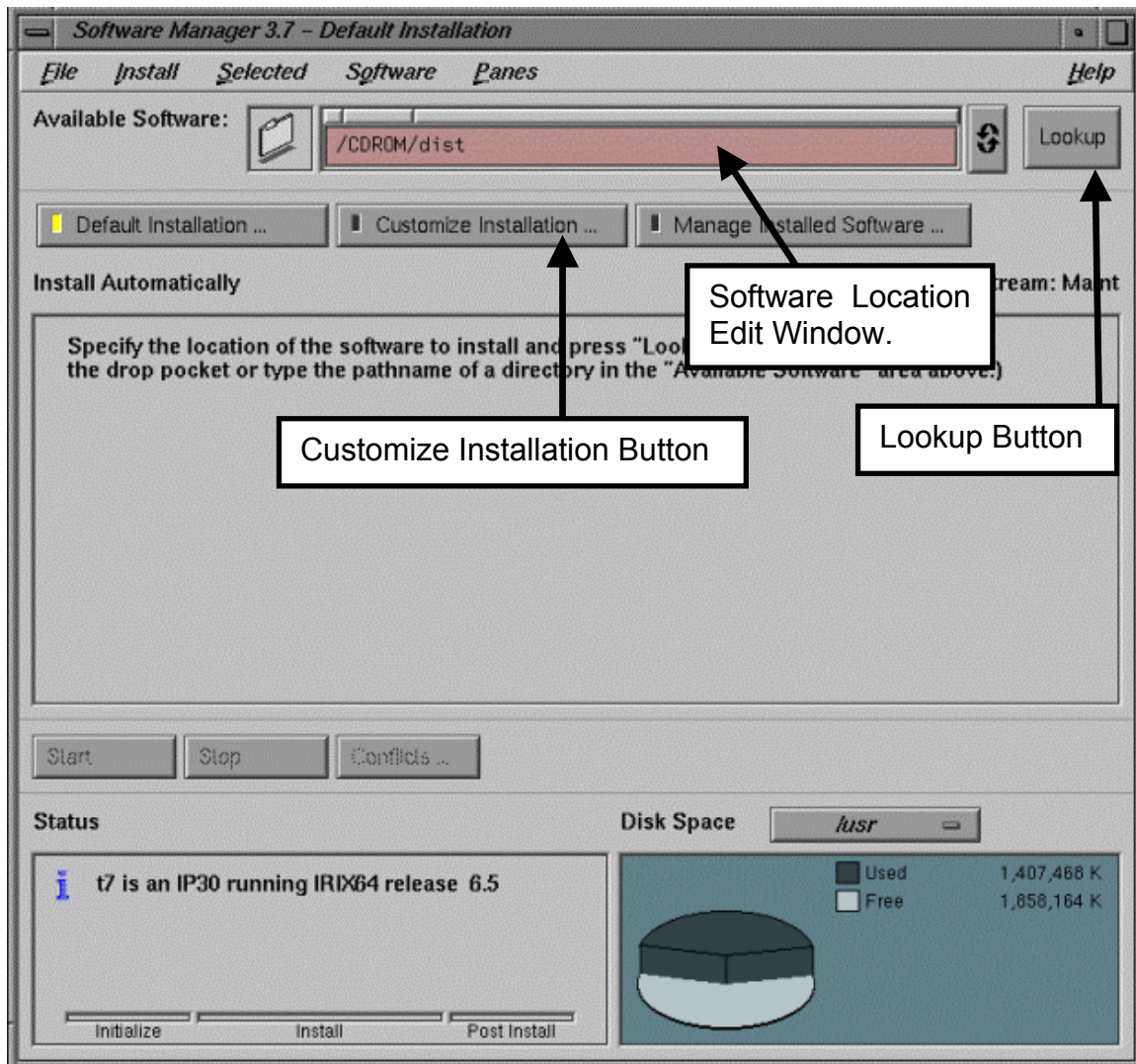
3. If there are no installed patches, complete Tech Pub 2329420 Rev 2 Signa Lx Release 9.0 Patch Software Load Instructions prior to loading Patch 4N.

3- LOADING SIGNA LX RELEASE PATCH FOR 9.0 M4 FROM CD-ROM

Note

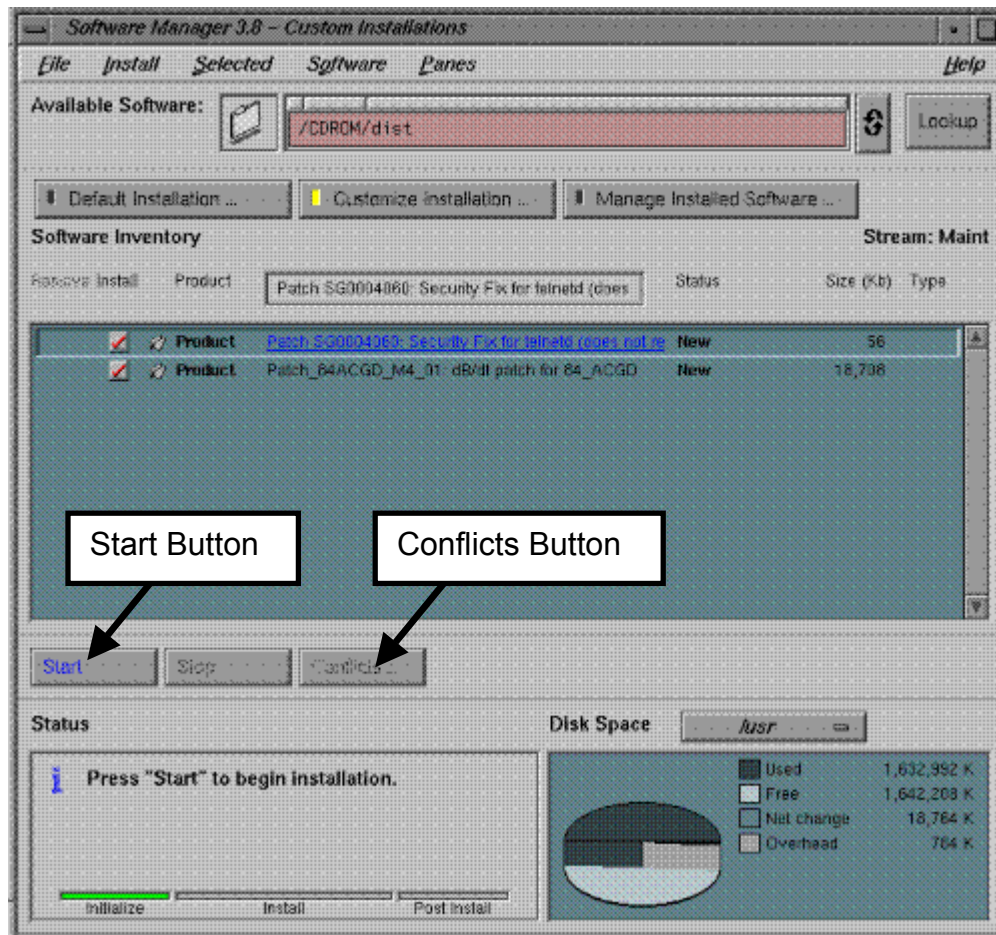
Do not use the Guided Install GUI when loading this patch CD. Changes need to be made using the Software Manager.

1. Make sure the system is idle (not scanning, filming, viewing or networking). The best way to confirm the system is idle is to restart the system and log in as root (in this case, proceed to step 3).
2. Inside a C Shell window, log in as the root user by typing the following:
su - <Enter>
Password: **operator <Enter>** ("operator" is the system default root password. Use whatever password is appropriate for the system you are working on if it has been changed).
3. Place the "Signa 9.0 M4 Patch" 2350782 Rev: SW001, CD in the system (not PC) CD drive. Wait 20 seconds for the system to see the CD until the light on the front of the CD drive stops blinking.
4. Inside the C Shell, type the following:
 - a) **mediad <Enter>** (This command mounts the CDRom).
 - b) **swmgr <Enter>** (This command starts the software manager which is used to load the patch).



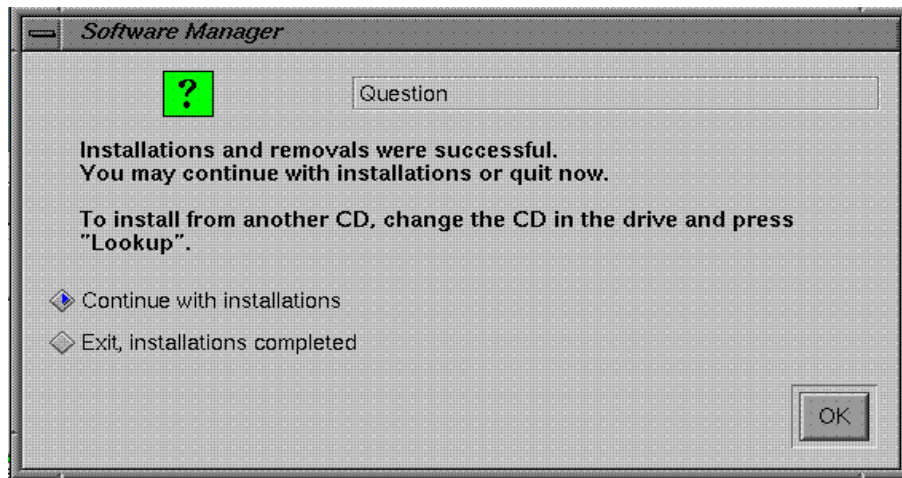
SOFTWARE MANAGER INITIALIZATION
ILLUSTRATION 3-1

- Once the Software Manager starts, begin typing the directory location where the patches are stored (case sensitive) in the **[Software Location Edit Window]**. See Illustration 3-1.
/CDROM (The system will automatically enter **"ldist"** after you type /CDROM)
- Click the **[Lookup]** button found to the right of the **[Software Location Edit Window]**. See Illustration 3-1.
- Click the **[Customize Installation]** button at the top of the screen. See Illustration 3-1.



CUSTOMIZE INSTALLATION SCREEN EXAMPLE
ILLUSTRATION 3-2

8. All available patches should be visible at this time in the selection window. Any patches that apply to your system and are not loaded will be selected (check marked). Verify the appropriate box is checked corresponding the system software version. Note: only the correct software version should be checked. See Illustration 3-2.
- ✓ **Product** Patch_90_M4_04: Tools SRFD2 Patch N for 90 M4 Release **Patch Upgrade**
9. If the **[Start]** button activates, proceed to 9.a. If the **[Conflicts]** button activates proceed to section 9.b
 - a) The **[Start]** button will activate (The letters in "Start" will change colors or shades). If the **[Start]** button activates, push it and the patch load will begin. A progress bar will open and count up from 0 to 100%. The load takes less than 1 minute. When the progress bar goes away, a Question window will be displayed. Select **[Exit, Installations completed]** and **[Ok]** to quit the Software manager. Proceed to step 10.
 - b) After the patch is selected, if there is a problem, the **[Conflicts]** button will activate, (the letters in "Conflicts" change colors or shades). If this happens, this indicates that your system does not need this patch and the indicated patch will not be loaded. Push the **[Conflicts]** to view the error if desired. Call the online center for support.
10. After installation, select Exit, Installations completed checkbox, and select **[Ok]**. See Illustration 3-3.



INSTALLATION DECISION WINDOW
ILLUSTRATION 3-3

11. In the C Shell type the following (case sensitive) to eject the CD-ROM from the drive:

umount /CDROM <Enter> (Un-mounts the CD from the system)
eject /CDROM <Enter> (Opens the CD drawer)

12. If the patches were required and loaded, verify they are installed by typing in the C Shell:

/usr/g/bin/getver <Enter> You should see an entry for the LX patch as follows
(for example only: the bold lines are related directly to this update):

Build number for MrpApps is 90.29F_M4_0139.z
Build number for PostSdC is 90.29F_M4_0139.z
Build number for cclass is 90.29F_M4_0139.z
Build number for driverSupport is 90.29F_M4_0139.z
Build number for insite is 2.3-mips
Build number for install is 90.29F_M4_0139.z
Build number for os_cd is 6_11.1.a
Build number for os_u_cd is 6u11.6.a

Build number for patch_90_M4_04 is 90.29F_M4_0139.z.PN

← The N indicates a
successful load of the
SRFD2 patch.

Build number for patch_90_M4_05 is 90.29F_M4_0139.z.PH
Build number for patch_90_M4_06 is 90.29F_M4_0139.z.PG
Build number for patch_90_M4_07 is 90.29F_M4_0139.z.PL
Build number for patch_90_M4_08 is 90.29F_M4_0139.z.PH
Build number for patch_90_M4_09 is 90.29F_M4_0139.z.PD
Build number for patch_90_M4_11 is 90.29F_M4_0139.z.PG
Build number for patch_90_M4_12 is 90.29F_M4_0139.z.PK
Build number for patch_90_M4_13 is 90.29F_M4_0139.z.PJ
Build number for patch_90_M4_14 is 90.29F_M4_0139.z.PL
Build number for patch_90_M4_15 is 90.29F_M4_0139.z.PJ
Build number for patch_90_M4_17 is 90.29F_M4_0139.z.PJ

Build number for pc is 90.29F_M4_0139.z
Build number for z is Upgrade
MR Software release: 9.0.0139z

4- CONFIGURE THE SYSTEM TO RUN WITH THE UPDATE

4-1 Change “RF amp type” Setting in MRConfig

Note:

The **Guided Install** Tool does not edit the RF amp type field in MRConfig, nor does it have the option for SRFD2. Use **Config File Manager** to change the setting or severe image quality problems may be seen.

1. Start **Config File Manager** under **[Utilities]** from the service desktop.
2. Select **[MR Config File]**.
3. Select the **RF Amplifier Type** config field.
4. Type **10 <ENTER>** to set the RF amp type for the SRFD2 cabinet.
5. Select **[Quit]**.
6. A confirmation window will open, select **[Yes]**.
7. In the file dialog box select **[Yes]**.
8. Reboot the system to apply the changes.

APPENDIX A - DETAILED PATCH DESCRIPTIONS

Note

These descriptions are excerpted from the readme.txt files that accompany each patch.

This readme file describes version N of patch_90_M4_04, which can be distributed through download. This patch contains all the changes necessary to support the SRFD2, plus the fix for fifteen problems found with the 90 'LEO1' release, thirteen of which have been previously addressed.

This patch is incompatible with patch_90_M4_03 - if a conflict with patch_90_M4_03 is detected, patch_90_M4_03 should be removed and replaced with patch_90_M4_10 or _14. This patch is compatible with patch_90_M4_10 and _14.

PROBLEM 1 - Unable to open HostVmSampleData.dat message
previously partially addressed in version D of this patch
previously fully addressed in version J of this patch

SPR - MRIge69010 Hackensack: Odd message ULHC in message box, won't go away

SPR - MRIge68800 Erronous mesg. Keeps appearing, but doesn't seem to cause prob

SPR - MRIge73895 "Unable to open HostVmSampleData.dat" error message on 90, 91

Description -

The vacuum monitor is defaulting to active on non-TRM systems. The error message is a nuisance only and does not cause any system malfunction. The resolution is to change the system start-up script, so as to start only on TRM.

Workaround - None.

The fix is in the following subsystems:

HOST: startmon script

PROBLEM 2 - Coil Config Manager does not work for ENDOATD and ATD3ENDO on 1.5T
previously addressed in versions D and J of this patch

SPR - MRIge71202 Errors in CoilConfig files prevent use of some coils in 9.0

Description -

The original 1.5T Coil Config files for Release 9.0 did not include information for two prostate spectroscopy coils: ENDOATD and ATD3ENDO. The patch includes coil config files for the two coils.

Workaround - None.

The fix is in the following subsystems:

HOST: Coil Config files

PROBLEM 3 - Cannot Archive or Network if Service or IIP Load done with Signa up
previously addressed in versions D and J of this patch

SPR - MRIge69757 insite/iip changed ownership on critical files

Description -

Unable to network images or archive images after a Restricted Service Load or IIP Load is done with Signa running. This is because critical files in sdc account had ownership changed to insite/iip, preventing user account sdc from accessing them as needed.

Workaround - Remove the soft link between the service directory and /tmp by typing:

```
rm /usr/g/csd/mr_csd/serviceDesktop/tmp
```

Then make sure the permissions on the /tmp directory are "drwxrwxrwt" and the owner is set to "sys".

The fix is in the following subsystems:

HOST: remove the soft link between /usr/g/csd/mr_csd/serviceDesktop/tmp and /tmp

PROBLEM 4 - Utilities menu covers start button on service desktop
previously addressed in versions D and J of this patch

SPR - MRIge71003 Utilities menu covers start button on service desktop

Description -

In the Service Desktop Manager (SDM), when one selects the Utilities Button, the horizontal scroll bar overlaps the START Button which is just below the Utilities Menu. To fix this problem, the height of the scrolled Window was reduced.

Workaround - Use the edge of the menu to select the desired choice.

The fix is in the following subsystems:

HOST: SDM

PROBLEM 5 - Scans on wrong grad coil after amp trip
previously addressed in versions D and J of this patch

SPR - MRIge65565 System scanning with wrong grad coil after amp trip

SPR - MRIge70699 Edison - FOV size is not what is prescribed

SPR - MRIge70001 System scanning with wrong grad coil after amp trip

Description -

Intermittently, for Twin systems only, after an amplifier trip, the scanner was scanning with the wrong gradient coil, resulting in images that appeared as if they were scanned at the wrong FOV. Additionally, sometimes MDS link errors 2225693, 2225699, and 2225704 are logged.

Workaround - Reboot Signa.

The fix is in the following subsystems:

Realtime: SPI

PROBLEM 6 - gamma lookup table lost after patch 4D
previously addressed inversion J of this patch

CQA - 1021834 9.0 Monitor, rungamma

SPR - MRIge72689 9.0 Monitor, rungamma

CQA - 1021027 9.0 LCD config, rungamma

SPR - MRIge72255 9.0 LCD config, rungamma

CQA - 1021111 9.0 Gamma tbl, rungamma

SPR - MRIge72321 9.0 Gamma tbl, rungamma

SPR - MRIge73227 startmon script corrupted - missing endif

The fix is in the following subsystems:

HOST: startmon script

PROBLEM 7 - patch 4D disable of archive/network
previously addressed inversion J of this patch

SPR - MRIge72918 Automation to enable archive / networking

The fix is in the following subsystems:

PATCH: installation script

PROBLEM 8 - ilinq gen2 lost after restore info previously addressed inversion J of this patch

SPR - MRIge72881 ILinq Gen2 needs save/restore patch

The fix is in the following subsystems:

HOST: save/restore scripts

PROBLEM 9 - system crashes running Noise Only Test
previously addressed inversion J of this patch

CQA - 10110845 ASPII-System crashes when running Noise Only Test

SPR - MRIge70673 ASPII-System crashes when running Noise Only Test

The fix is in the following subsystems:

HOST: noise_script

PROBLEM 10 - unable to edit 1st coil in site list using CFM
previously addressed inversion J of this patch

CQA - 1020025 LEO1 Coilconfig name 4SHOULDR

SPR - MRIge71834 LEO1 Coilconfig name 4SHOULDR

SPR - MRIge71974 Unable to Edit 1st coil in Site list using ConfigFile Manager

The fix is in the following subsystems:

HOST: Config File Editor

PROBLEM 11 - ucerd type not set correctly from Guided Install
previously addressed inversion J of this patch

SPR - MRIge72567 hardware_write script needs to be modified

The fix is in the following subsystems:

HOST: Guided Install scripts

PROBLEM 12 - shutdown hangs
previously addressed inversion J of this patch

CQA - 1022320 9.0 shutdown hangs

SPR - MRIge73081 9.0 shutdown hangs

CQA - 1020064 LEO1 Shutdown hangs

SPR - MRIge71851 LEO1 Shutdown hangs

CQA - 1020398 9.0 no shutdown from tool menu

SPR - MRIge72050 9.0 no shutdown from tool menu

CQA - 1022213 9.0 shutdown not complete[D

SPR - MRIge73002 9.0 shutdown not complete[D

CQA - 1022606 9.0 shutdown hangs

SPR - MRIge73310 9.0 shutdown hangs

The fix is in the following subsystems:

HOST: stopmon script

PROBLEM 13 - LV shim protocol has APS peak on water instead of peak previously addressed inversion J of this patch

SPR - MRIge73790 LV shim protocol has APS peak on water instead of peak

Description -

The first scan protocol under LVshim is a localizer and it has peak selected for APS. The second series is the one that you actually scan and run the analysis on. This has APS peak on water. This is wrong and should be changed to PEAK.

Workaround - Manually edit the protocol.

The fix is in the following subsystems:

Protocol: LVshim

PROBLEM 14 - Table does not move fully to scan plane

SPR - MRIge74897 9.0 SmartPrep does not work

CQA - 1024575 9.0 SmartPrep does not work

Description -

This change to the Scan Room Interface fixes several issues with hardware in which the table does not make it all the way to the scan plane.

Workaround -

The fix is in the following subsystems:

REALTIME: sri executables

PROBLEM 15 - Intermittent PSD download failures

SPR - MRIge74287 Packet loss and associated download failures

SPE - MRIge67417 PSD time out errors due to lack of file pointers

Description -

PSD download failures occasionally seen at 9.0 sites

Workaround -

The fix is in the following subsystems:

REALTIME: ipg.o